

Nathan Dinh
CS-174 Section 02 – Server Web Programming
Spring 2025

HW6 – Stock Search Web App – Flask + Tiingo API

=====

Deployed EC2 URL:

<http://ec2-44-247-221-104.us-west-2.compute.amazonaws.com/>

Flask Deployed URL:

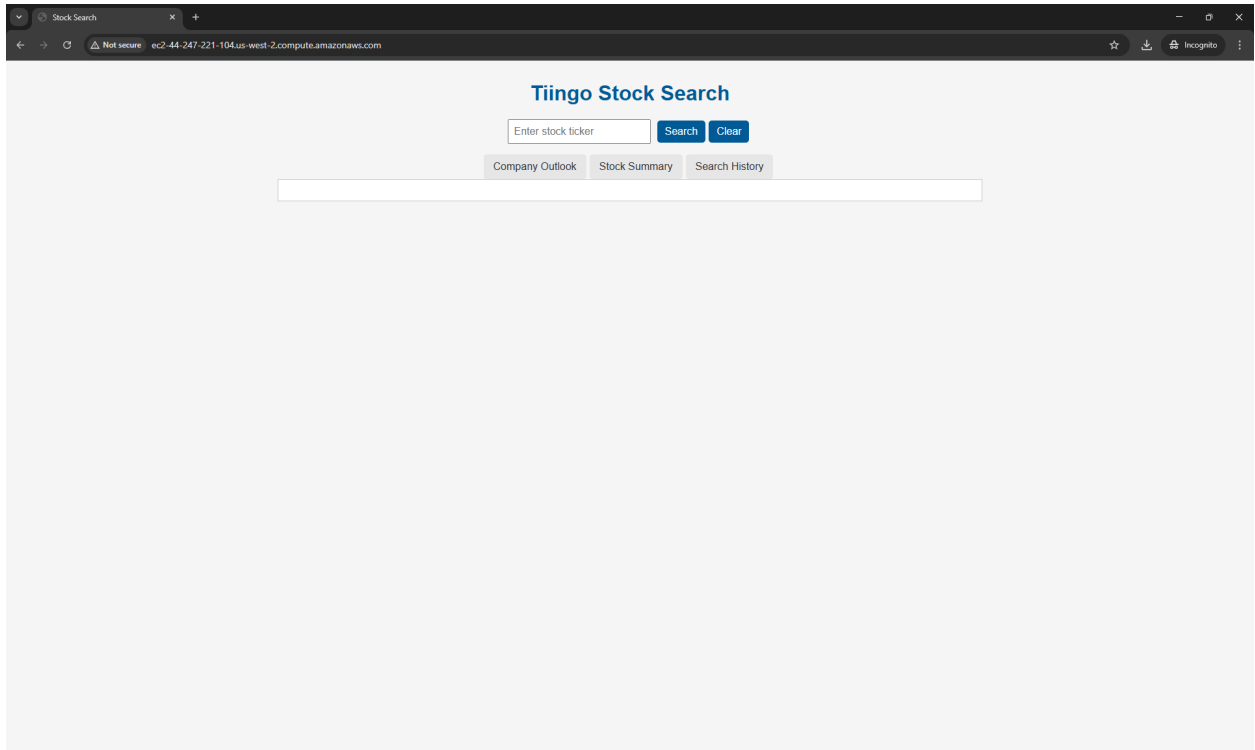
<http://ec2-44-247-221-104.us-west-2.compute.amazonaws.com:8000>

Description:

This web app allows users to search for stock ticker symbols and retrieve real-time company and stock data using the Tiingo API. It includes caching, search history, and a clean user interface using HTML, CSS, and JavaScript.

Main Features:

- Search stock by ticker (e.g., AAPL)
- View company details and stock summary
- Last 10 searches saved in SQLite
- Results cached for 15 minutes (**EXTRA CREDIT**)
- Visual Message “Served from cache” (**EXTRA CREDIT**)
- Tabbed interface with arrow icons for price change
- Input validation and error messages



Home Page

Stock Search

Not secure ec2-44-247-221-104.us-west-2.compute.amazonaws.com Incognito

Tiingo Stock Search

Search

Clear

Company Outlook

Stock Summary

Search History

Company Name	NVIDIA Corp
Ticker	NVDA
Exchange	NASDAQ
Start Date	1999-01-22
Description	NVIDIA is the world leader in accelerated computing....

Search with “NVDA” ticker

Stock Search

Not secure ec2-44-247-221-104.us-west-2.compute.amazonaws.com

Incognito

Tiingo Stock Search

Search

Clear

Company Outlook

Stock Summary

Search History

Ticker	NVDA
Date	2025-05-01
Prev Close	108.92
Open	113.08
High	114.94
Low	111.3
Last	null
Change	-108.92 ↓
Change %	-100.00% ↓
Volume	236121507

Show results stock summary

Stock Search

Not secure ec2-44-247-221-104.us-west-2.compute.amazonaws.com Incognito

Tiingo Stock Search

Search

Clear

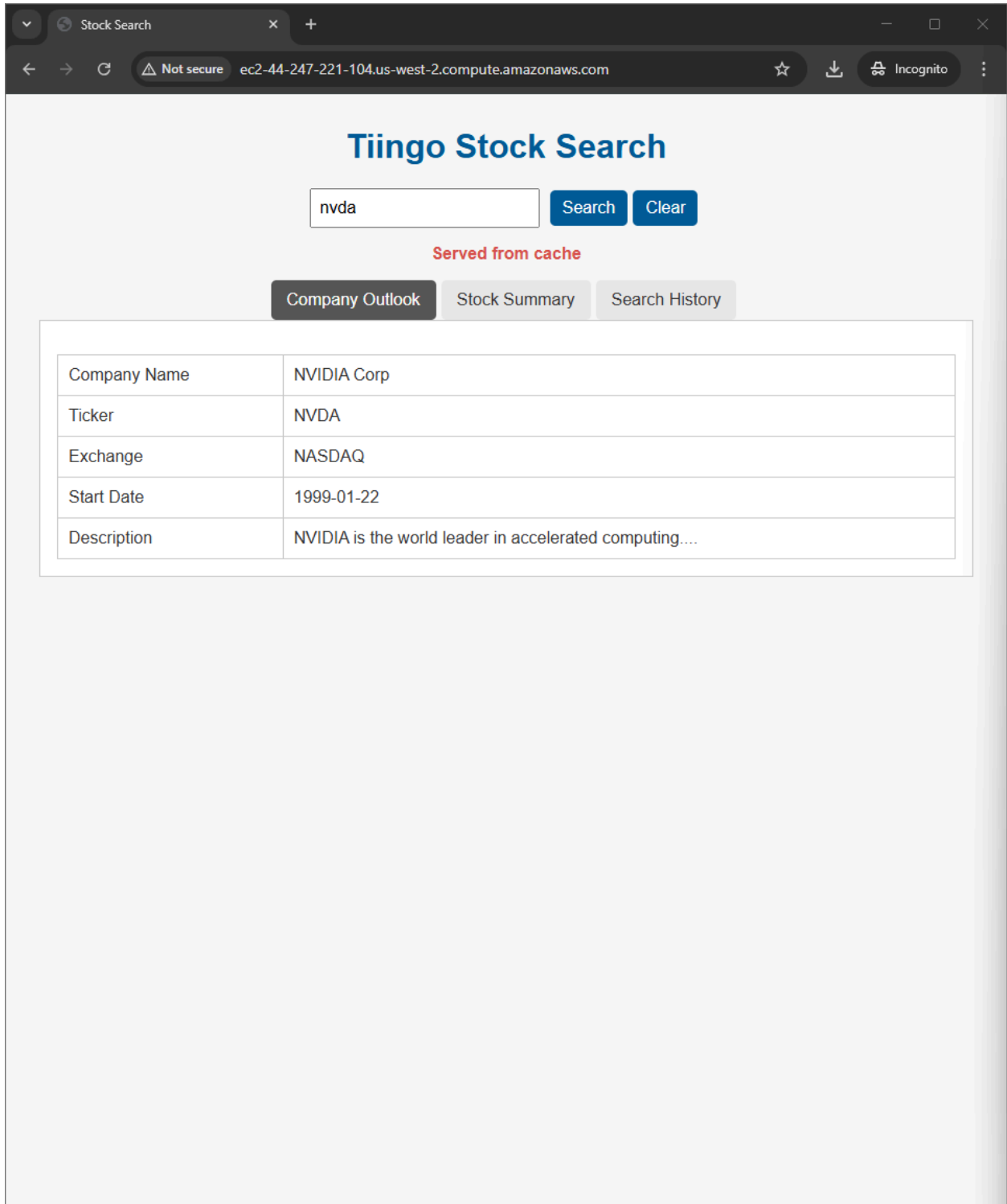
Company Outlook

Stock Summary

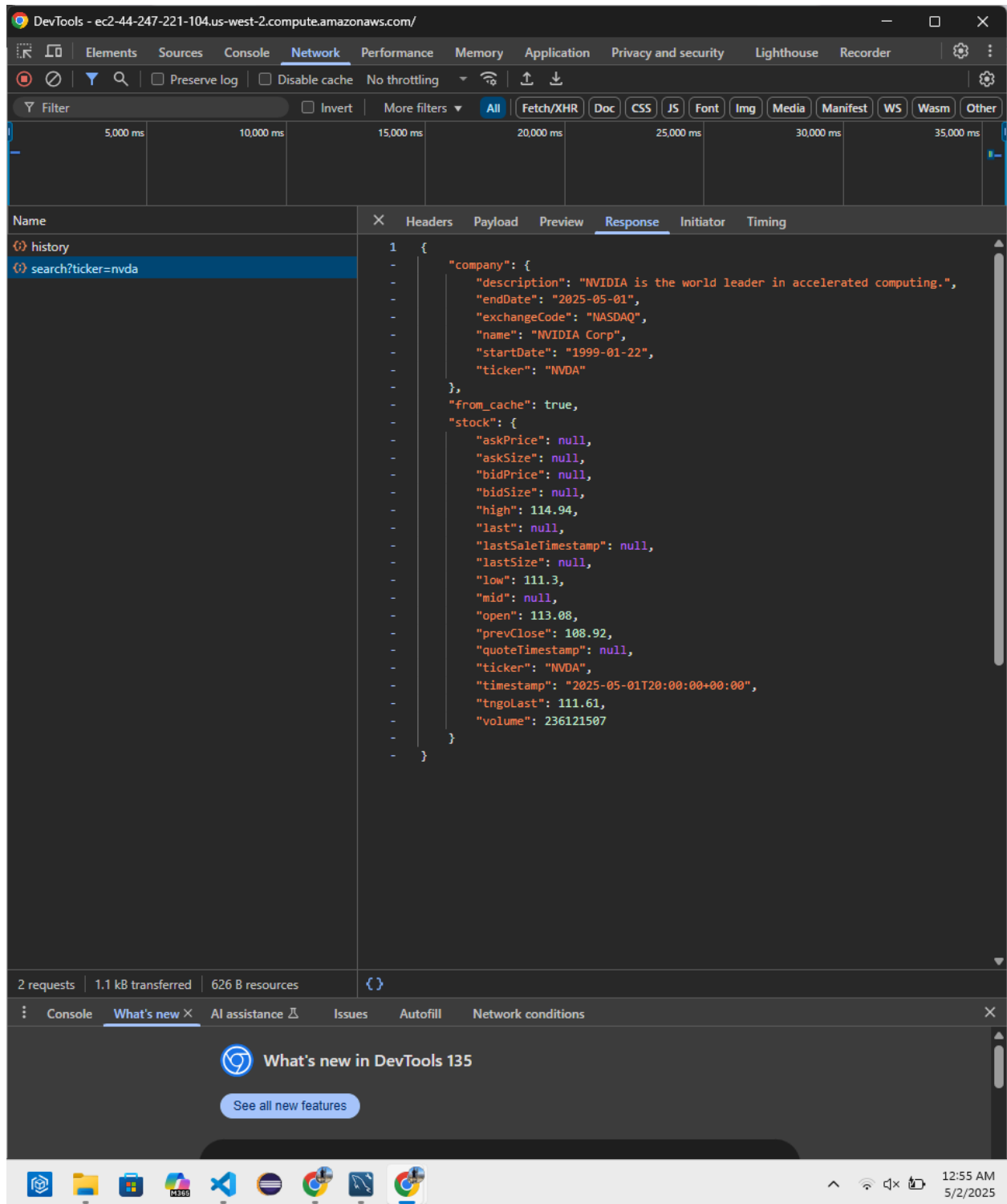
Search History

Ticker	Timestamp
NVDA	2025-05-02 07:53:55
AAPL	2025-05-02 07:47:30

Show Search History



If I search again in 15 minutes, it will get data from cache (database) instead



Response from API call with "from_cache" attribute is true

Stock Search

Not secure ec2-44-247-221-104.us-west-2.compute.amazonaws.com

Incognito

Tiingo Stock Search

SearchClear

Company Outlook

Stock Summary

Search History

Ticker	BNGO
Date	2025-05-01
Prev Close	3.99
Open	3.94
High	4.12
Low	3.91
Last	null
Change	-3.99 ↓
Change %	-100.00% ↓
Volume	37594

Search another ticker name “BNGO”

Stock Search

Not secure ec2-44-247-221-104.us-west-2.compute.amazonaws.com

Incognito

Tiingo Stock Search

SearchClear

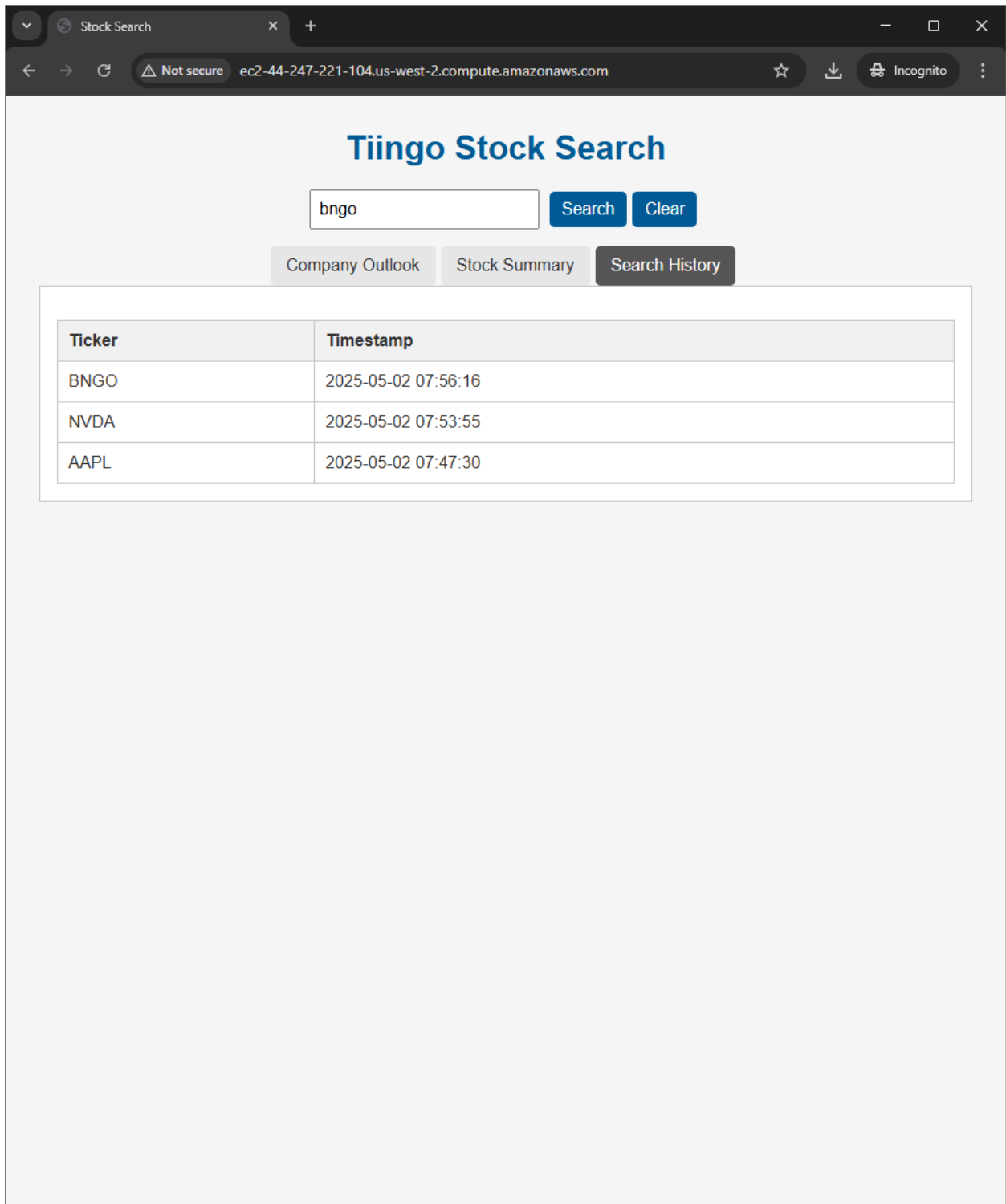
Company Outlook

Stock Summary

Search History

Company Name	Bionano Genomics Inc
Ticker	BNGO
Exchange	NASDAQ
Start Date	2018-09-21
Description	Bionano Genomics is a provider of genome analysis solutions that can enable researchers and clinicians to reveal answers to challenging questions in biology and medicine. The Company's mission is to transform the way the world sees the genome through OGM solutions, diagnostic services and software. The Company offers OGM solutions for applications across basic, translational and clinical research. Through its Lineagen, Inc. d/b/a Bionano Laboratories business, the Company also provides diagnosti...

Show results as company outlook



Update search history

The screenshot shows the Chrome DevTools Network tab with the 'Response' sub-tab selected. The left pane lists several requests, with 'search?ticker=bngo' highlighted. The right pane displays the JSON response for this request. The response includes company information and stock data. The 'from_cache' attribute is set to false, indicating a fresh fetch from the Tiingo API.

```
1 {
  -   "company": {
  -     "description": "Bionano Genomics is a provider of genome analysis solutions",
  -     "endDate": "2025-05-01",
  -     "exchangeCode": "NASDAQ",
  -     "name": "Bionano Genomics Inc",
  -     "startDate": "2018-09-21",
  -     "ticker": "BNGO"
  -   },
  -   "from_cache": false,
  -   "stock": {
  -     "askPrice": null,
  -     "askSize": null,
  -     "bidPrice": null,
  -     "bidSize": null,
  -     "high": 4.12,
  -     "last": null,
  -     "lastSaleTimestamp": null,
  -     "lastSize": null,
  -     "low": 3.91,
  -     "mid": null,
  -     "open": 3.94,
  -     "prevClose": 3.99,
  -     "quoteTimestamp": null,
  -     "ticker": "BNGO",
  -     "timestamp": "2025-05-01T20:00:00+00:00",
  -     "tnoLast": 4.0,
  -     "volume": 37594
  -   }
  - }
```

5 requests | 5.0 kB transferred | 3.8 kB resources

What's new in DevTools 135
[See all new features](#)

API response since it's the first time to search this ticker so there is no cache data, fetch from Tiingo API. You can see the "from_cache" attribute here is false

Stock Search

Not secure ec2-44-247-221-104.us-west-2.compute.amazonaws.com

Incognito

Tiingo Stock Search

Search

Clear

Served from cache

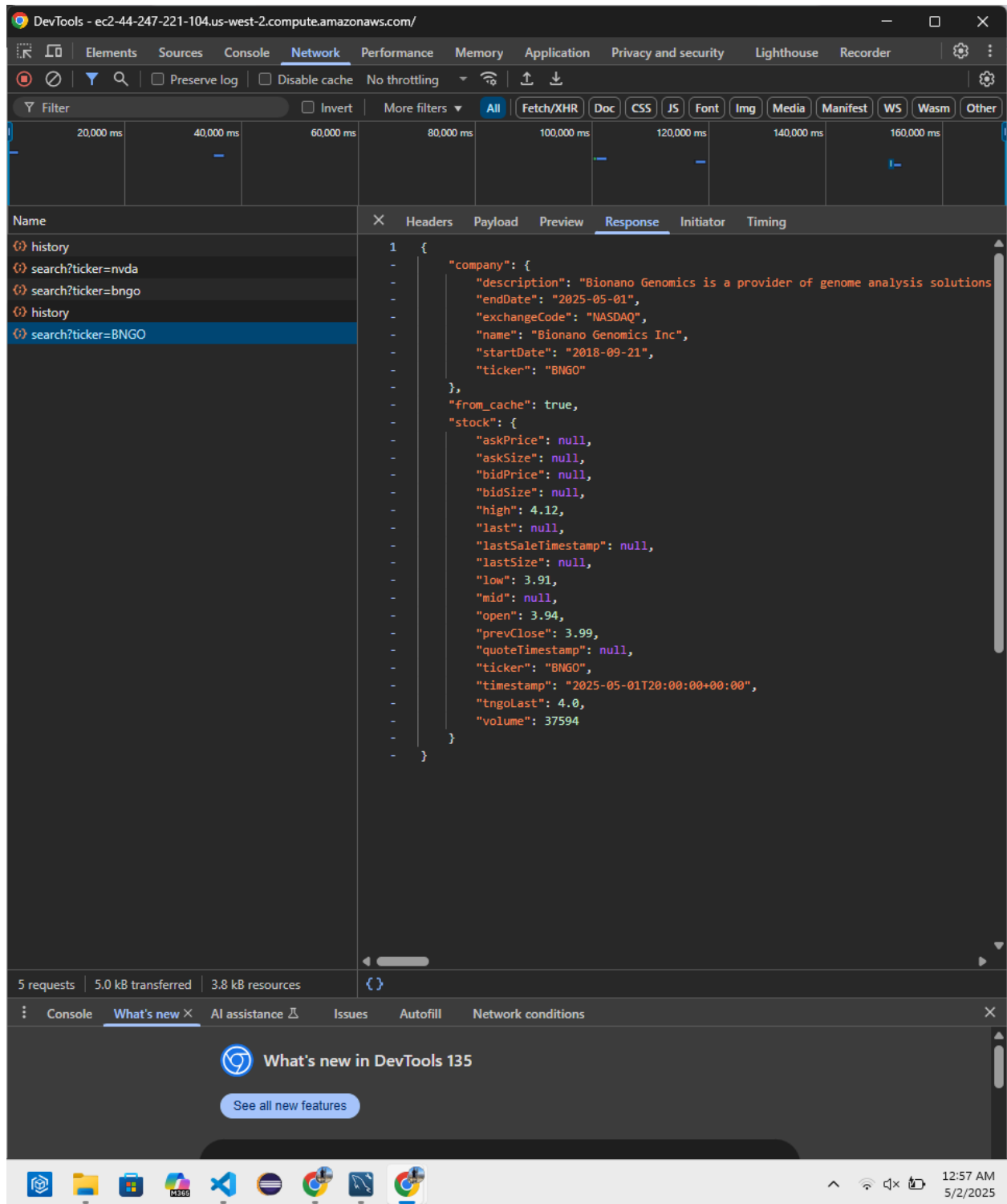
Company Outlook

Stock Summary

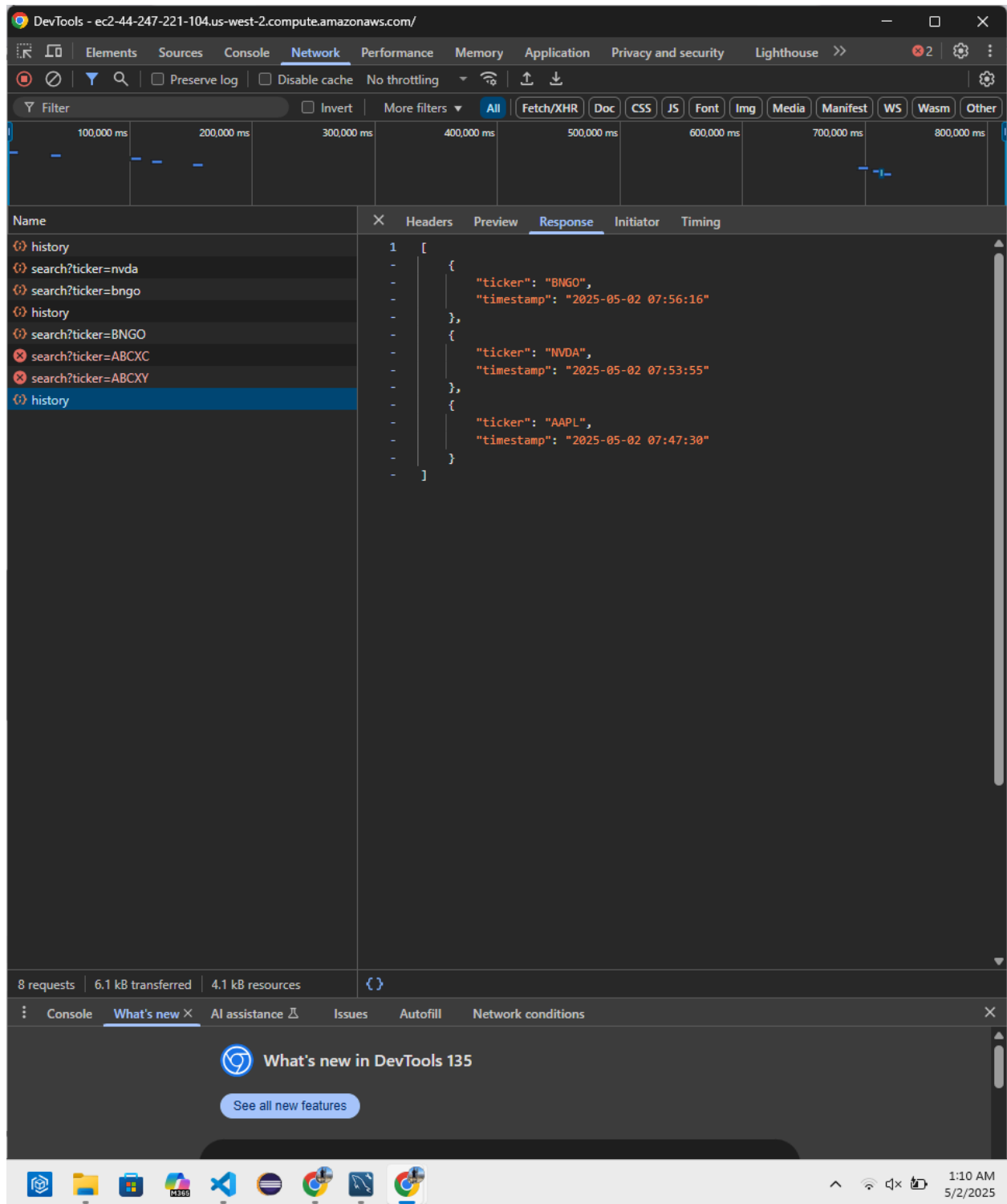
Search History

Company Name	Bionano Genomics Inc
Ticker	BNGO
Exchange	NASDAQ
Start Date	2018-09-21
Description	Bionano Genomics is a provider of genome analysis solutions that can enable researchers and clinicians to reveal answers to challenging questions in biology and medicine. The Company's mission is to transform the way the world sees the genome through OGM solutions, diagnostic services and software. The Company offers OGM solutions for applications across basic, translational and clinical research. Through its Lineagen, Inc. d/b/a Bionano Laboratories business, the Company also provides diagnosti...

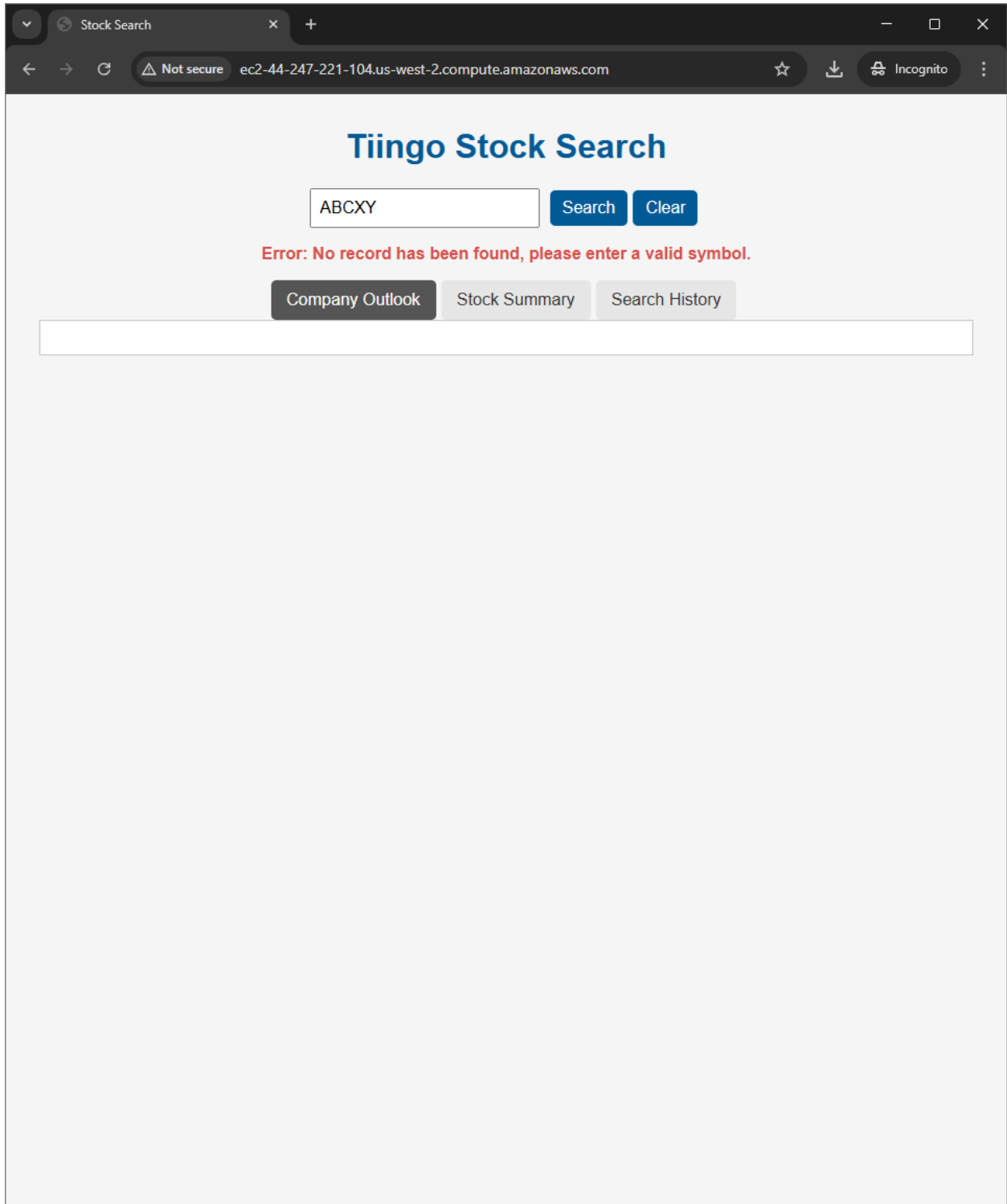
If I search this again not in 15 mins then it show “Served from cache” and the data is getting from cache



Here is API response with data from cache



API Response from GET history request



Return message as “Error: No record has been found, please enter a valid symbol.” if the ticker is not found.